Vacuum-Compatible Compact Linear Stage

XY(Z) Combinations with DC Gear Motor, Various Travel Range Variants



M-110 • M-111 • M-112 V6

- Suitable for vacuum up to 10^{-6} hPa
- Very compact due to folded drive
- Closed-loop DC motors
- Travel ranges: 5, 15, and 25 mm
- Noncontact reference and limit switches
- XY setups can be mounted directly

Compact linear stage, versatile in use
Linear stage with compact size due to folded drive with belt drive. Clear aperture with 10 mm Ø. XY setups can be mounted directly without adapter plate.

Reduced outgassing especially for use in vacuum environments
Most notably, for the lubrication of the guides and the spindle a material with particularly low outgassing is used. This allows a bakeout temperature of up to 80° C. With this and other modifications of the PI V6 vacuum class, this positioner can be used up to 10^{-6}hPa.

Drive
- Closed-loop DC gear motor
- Backlash-compensated threaded spindle

Position measuring
- Rotary encoder on the motor shaft
- Noncontact Hall effect reference and limit switches

Application fields
Vacuum chambers. Research. Beamline instrumentation. Microscopy. Electronics assembly and inspection.

Motion	Unit		M-110.1VG1	M-111.1VG1	M-112.1VG1
Active axes			X	X	X
Travel range in X	mm		5	15	25
Maximum velocity in X, unloaded	mm/s		1	1.5	1.5

Positioning	Unit	Tolerance	M-110.1VG1	M-111.1VG1	M-112.1VG1
Minimum incremental motion in X	µm	Typ.	0.05	0.05	0.05
Unidirectional repeatability in X	µm	Typ.	±0.125	±0.125	±0.125
Bidirectional repeatability in X	µm	Typ.	4	4	4
Backlash in X	µm	Typ.	3	3	3
Reference switch			Hall effect	Hall effect	Hall effect
Limit switches			Hall effect	Hall effect	Hall effect
Motor encoder			Incremental rotary encoder	Incremental rotary encoder	Incremental rotary encoder
Motor encoder: Sensor signal			A/B quadrature, RS-422	A/B quadrature, RS-422	A/B quadrature, RS-422
Resolution, motor encoder	Cts./rev.		2048	2048	2048

Drive Properties	Unit	Tolerance	M-110.1VG1	M-111.1VG1	M-112.1VG1
Drive type			DC gear motor	DC gear motor	DC gear motor
Nominal voltage	V		12	12	12
Nominal current, RMS	A	Typ.	0.16	0.32	0.32
Drive force in positive direction of motion in X	N	Typ.	10	10	10
Drive force in negative direction of motion in X	N	Typ.	10	10	10

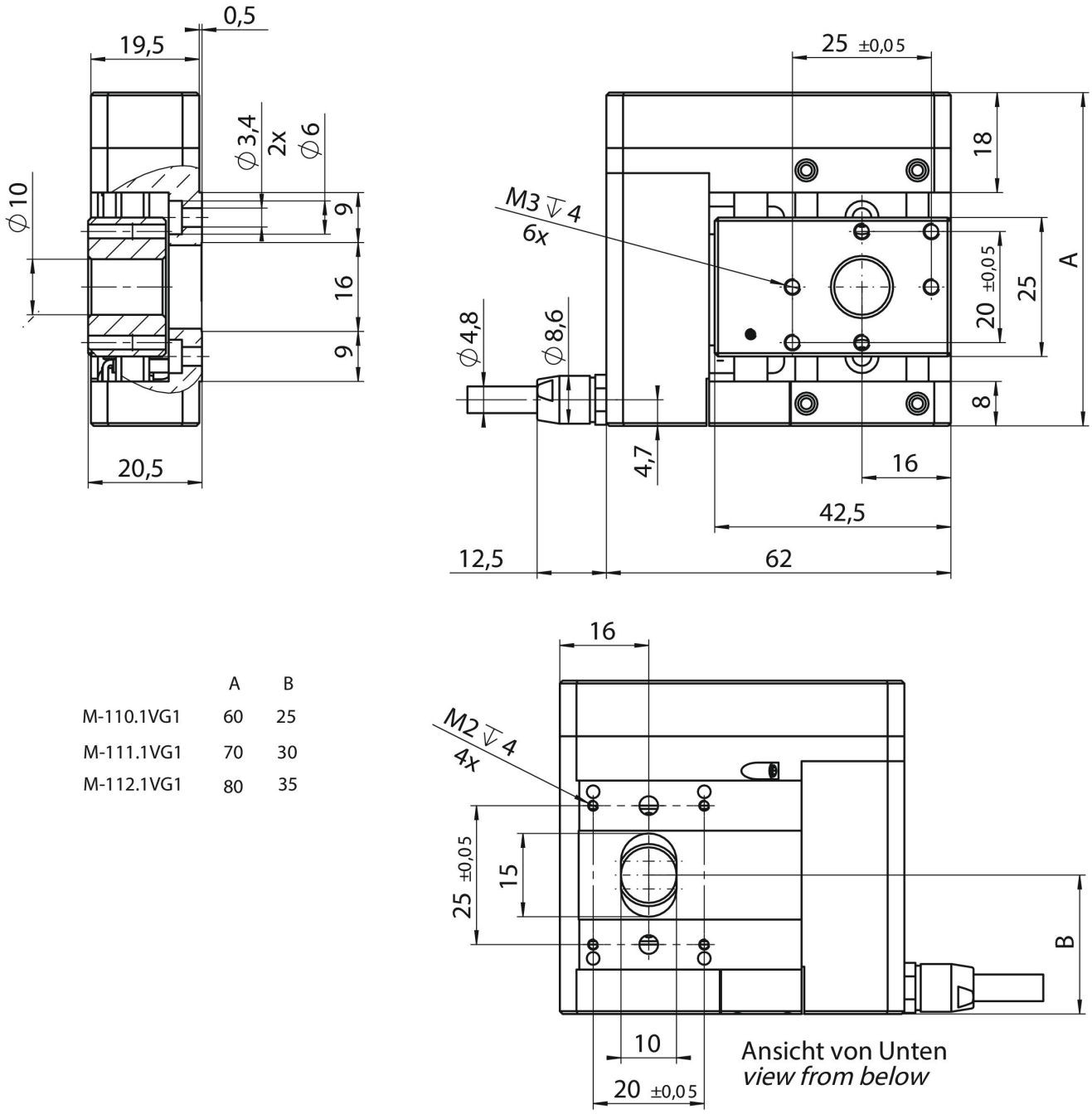
Mechanical Properties	Unit	Tolerance	M-110.1VG1	M-111.1VG1	M-112.1VG1
Permissible push force in Y	N	Max.	15	10	10
Permissible push force in Z	N	Max.	30	30	20
Permissible torque in θX	N·m	Max.	0.15	0.15	0.15
Permissible torque in θY	N·m	Max.	0.15	0.15	0.15
Permissible torque in θZ	N·m	Max.	0.15	0.15	0.15
Holding force in X, passive	N		10	10	10
Moved mass in X, unloaded	g		11	11	11
Drive screw type			Fine-threaded spindle	Fine-threaded spindle	Fine-threaded spindle
Drive screw pitch	mm		0.4	0.4	0.4
Gear ratio i			29791 : 729	29791 : 729	29791 : 729
Guide			Ball guide	Ball guide	Ball guide
Overall mass	g		250	270	300
Material			Anodized aluminum	Anodized aluminum	Anodized aluminum

Miscellaneous	Unit		M-110.1VG1	M-111.1VG1	M-112.1VG1
Operating temperature range	°C		-20 to 65	-20 to 65	-20 to 65
Vacuum class	hPa		10^{-6}	10^{-6}	10^{-6}
Connector			HD D-sub 26 (m)	HD D-sub 26 (m)	HD D-sub 26 (m)
Cable length	m		0.5	0.5	0.5
Recommended controllers / drivers			C-863 C-884 G-901 G-910 ACS modular controller	C-863 C-884 G-901 G-910 ACS modular controller	C-863 C-884 G-901 G-910 ACS modular controller

Note on minimum incremental motion: Values are achieved with the recommended controller
Note on nominal current: Values apply per phase

At PI, technical data is specified at 22 ±3 °C. Unless otherwise stated, the values are for unloaded conditions. Some properties are interdependent. The designation "typ." indicates a statistical average for a property; it does not indicate a guaranteed value for every product supplied. During the final inspection of a product, only selected properties are analyzed, not all. Please note that some product characteristics may deteriorate with increasing operating time.

Drawings / Images



	A	B
M-110.1VG1	60	25
M-111.1VG1	70	30
M-112.1VG1	80	35

M-11x.1VG1, dimensions in mm. Note that a comma is used in the drawing instead of a decimal point.

Order Information

M-110.1VG1
Vacuum-compatible compact linear stage; DC gear motor; 5 mm travel range; 30 N load capacity; 1 mm/s maximum velocity; fine-threaded spindle; incremental rotary encoder, 2048 cts./rev. resolution, A/B quadrature, RS-422; vacuum compatible to 10^{-6} hPa; 0.5 m cable length

M-111.1VG1
Vacuum-compatible compact linear stage; DC gear motor; 15 mm travel range; 30 N load capacity; 1.5 mm/s maximum velocity; fine-threaded spindle; incremental rotary encoder, 2048 cts./rev. resolution, A/B quadrature, RS-422; vacuum compatible to 10^{-6} hPa; 0.5 m cable length

M-112.1VG1
Vacuum-compatible compact linear stage; DC gear motor; 25 mm travel range; 20 N load capacity; 1.5 mm/s maximum velocity; fine-threaded spindle; incremental rotary encoder, 2048 cts./rev. resolution, A/B quadrature, RS-422; vacuum compatible to 10^{-6} hPa; 0.5 m cable length